



MA 606: Mathematical Models for Infectious Diseases

Course Code: MA 606

Course Name: Mathematical Models for Infectious Diseases

L-T-P-C: 3-0-0-3

Approval: One-time approval recommended by the 26th BoA and approved by the 18th Senate (approved earlier in the 2nd adhoc Senate meeting as MS 650)

Course Content

Over the last two decades, mathematical models have seen a huge development in all aspects of infectious diseases, from microbiology to epidemiology and evolution. The programme covers introductory and advanced concepts in mathematical modelling of infectious diseases, including:

- Mathematical review (calculus, probabilities ...)
- Deterministic and stochastic models
- Network analysis
- Within-host dynamics of viral and bacterial infections
- Applied programming with R
- Statistical data modelling
- Computer-based simulations